CAISO OASIS API — Connection & Usage Spec

What this system is

OASIS (Open Access Same-Time Information System) is CAISO's public data system for market and grid data — prices, bids, load, outages, and reference/resource information. Unlike CAISO's other systems, OASIS doesn't require certification or a market participant account to access — it's open to the public. No login, no approval process, no credentials needed for what we're pulling.

What's in scope for us

Two categories of data matter for the bid-gap analysis:

- **Market/operational data** (clearing prices, load, resource info): near-real-time, no access delay.
- **Public Bid Data** (what a resource actually submitted): confidential until 90 days after the trading date, then becomes public. This delay is a fixed disclosure rule, not a data-processing lag —

there's no faster legitimate path to it, certified or not.

Identity resolution — a prerequisite, not a data pull itself

OASIS keys everything by **Resource ID / PNode**, not plant name. Before any plant-specific data can be requested, its Resource ID needs to be resolved via CAISO's Master File / resource listing. This is a one-time (or infrequently-refreshed) lookup per plant, not something to redo on every pull.

How requests work, conceptually

Data is requested via URL-based queries against a single-report endpoint. Each request needs, at minimum:

- Which report is being requested (each dataset — clearing prices, public bids, resource listings, etc. — has its own report identifier)
- A date/time range
- Which market (day-ahead vs. real-time) the data applies to, where relevant
- Which resource/node the data applies to, where relevant

Responses can be returned in either XML (the default) or CSV (via a format flag) — CSV is generally easier to work with for tabular analysis.

Report identifiers and version numbers are not stable long-term — CAISO has revised the OASIS interface specification multiple times, and report names/versions have changed across releases. Whatever identifier gets used for Public Bid Data specifically should be confirmed against the *current* OASIS Interface Specification (published on CAISO's site) rather than hardcoded from an older reference.

Constraints that shape how a pull needs to be structured

- **Date range per request is capped, and the cap varies by report.** Most operational reports allow up to 31 days per request. Public Bid Data specifically is more restrictive — one trading day per request. This asymmetry matters: a year of clearing-price data is a handful of requests, while a year of bid data for one resource is on the order of hundreds of requests.
- **Rate limiting exists.** CAISO throttles request frequency (on the order of one request every few seconds); exceeding it returns a "too many requests" error rather than a silent failure. Any automated pull needs deliberate pacing between

requests, and needs to handle a throttling response by backing off and retrying rather than treating it as a hard failure.

- **No data returned isn't always an error** — it can also mean the query's scope (too many nodes/variables at once) needs narrowing.

Existing tooling worth checking before building raw queries

An open-source Python package (`gridstatus`) already wraps many of CAISO's OASIS reports — clearing prices, load, curtailment, and others — handling the underlying query construction and response parsing. It's worth checking whether it covers a given report before writing a raw query against the API directly. Public Bid Data specifically may not be one of the datasets it wraps, given how infrequently that report is used relative to price/load data — that's worth confirming rather than assuming either way.

Manual/interim access (no code required)

Two browser-based options exist for pulling data without writing a query:

- The OASIS web interface itself, for report-by-report manual queries and downloads.
- A separate bulk historical downloader tool CAISO provides, oriented toward pulling larger historical ranges through a form-based interface rather than hand-built URLs.

Either is useful for spot-checking a plant's data by hand before automating against it.

Open items

- Confirm the current Public Bid Data report identifier and version against the live OASIS Interface Specification before building against it.
- Confirm whether `gridstatus` (or another existing wrapper) covers Public Bid Data, or whether that report needs a direct query.
- Decide how the daily incremental pull should handle a missed day (e.g., the job doesn't run for a stretch) — catch up automatically, or treat each day independently and accept gaps.